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Manuscript title: Review of Development in Air Quality Modeling and Air Quality Dispersion Models

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Abstract

Air dispersion models are the mathematical tools used for simulating the physical and chemical process governing the diffusion and transformation of pollutant in the atmosphere. The simplest dispersion models are steady-state Gaussian-plume models, they are based on mathematical approximation of plume behaviour and follows some basic assumptions that may not always present a realistic scenario. Despite having these limitations, they provide reasonable results when used aptly. More recently, advanced dispersion models are being developed, which are based on a more refined approach of simulating the dispersion phenomenon following the properties of the atmosphere rather than relying on general mathematical approximation. This has expanded the field of modelling to tackle difficult situations such as complex terrain and long-distance transport. In this review paper, the development in air quality modelling with emphasis on dispersion modelling has been presented. Further, few models are selected representing different categories in dispersion modelling, which are Gaussian models – AERMOD, CALINE4, Airviro Gauss, CTDMPLUS and FDM; Eulerian models – CALGRID, FARM and PANACHE; Lagrangian – GRAL, FLEXPART, AUSTAL2000 and HYSPLIT; Advanced Dispersion Models - UK-ADMS5, TAPM and CALPUF. A comparison has been done based on certain characteristic features obtained from various literature.

1. Introduction

Tracing back the development of air quality models leads us to the phase when major advancement was taking place around the world. Air quality models are the results of contributions of different ingenious minds which were looking for the solutions of different problems engulfing the developing nations, the fundamental being, the need to understand the dispersion of poisonous gas for military purpose, smoke clearing, tracking the path of radioactive pollutants after nuclear disasters, effects of pollutants on human health released from growing industries and vehicles on road. The ground-breaking development in models was done with the coming up of EPA in 1970. The primary characteristics of these models is the relationship between transport, dispersion and fate of a pollutant which follows a first order approximation.

The timeline for modelling development started from mid-1940. The first phase models were steady state with first order chemical reactions. During this phase, the focus of the academicians and researchers were on the single source. These models were effectively used for weapon testing and designing of stacks (Weber, 1982). The second phase focused on dispersion and transport of pollutant as a function of time with more advanced chemistry and parameter specifications. This period marked the development of multi-source dispersion models suitable for changing society from rural to urban. These models are simulated for estimating various criteria pollutants like SO₂, NO₂, and Particulate matter. The third phase is more recent and mark the new generation of models with complex chemical processes intermingling various phenomenon and interaction between parameters through feedback and

loop systems (Irwin, 2004).

2. Literature review

The documentation of the historical development gets emphasis with stern's work on review of various aspects of air pollutants, their effect, monitoring and surveillance, control and quality management, that has been spread across five volumes (Stern, 1976-77). The basic principle in modelling that is simulation of transportation and fate of pollutants was studied by many researchers of the time like DeMarrais work was focused on the meteorology of air pollution (DeMarrais, 1974). Turner's highlighted the historical perspective of dispersion modelling marking the fact that although effect of meteorology on dispersion was recognized long before, but research was done effectively after World War I (Turner, 1979). Paolo Zanetti in his book Air Pollution Meteorology described planetary boundary layer where vertical exchange of heat, momentum and moisture in atmosphere takes place (Zannetti, 1990). With the changing atmospheric scenarios different streams in air quality modelling developed whose emphasis was on the particular problem pertaining to air pollution. For modelling of buoyant emission, plume rise models were developed to measure the height of plume rise before stabilizing, meteorological models characterize the atmospheric properties, Dispersion models simulate the path of pollutant as it travels downwards, Chemistry models describes the chemical transformation (Irwin, 2004).

With the inception of EPA, many review articles were published co-relating the modelling development with Environmental Protection Agency's expansion, the major one was of National Research council describing the legislative and modelling tools for Ozone

(National Research Council, 1991), revision of modelling system to accommodate concentration level and human exposure to environment (National Research Council, 1992). Initially Air Quality Models were developed for planning purpose then with changing needs the focus has been shifted to industrial requirements. With the coming up of (AEC) 1954, modelling was taken up for impact assessment. In 1963 Clean Air Act was passed in US Which standardized the criteria for individual pollutants (Irwin, 2004). In 1970 Clean Air Act was amended and National Ambient Air Quality Standards were fixed for 6 pollutants: Carbon monoxide, total suspended particulate matter, Sulphur dioxide, oxidants, nitrogen dioxide and lead. The amendment of 1977 in clean air act formalize the use of dispersion modelling for air quality prediction. In early 1970s research on photochemical modelling has also become prominent, UAM-V (Urban Airshed Model Variable Grid) photochemical modelling system was the pioneering effort which had been widely used by EPA for air quality studies focussed on ozone. Seinfeld has done exhaustive research in photochemical modelling. He stated that the study in photochemical modelling started in 1969 when four different agencies started working simultaneously in this field: System Development Corp., General Research Corp., Systems, Science and Software, and California Institute of Technology/ Systems Applications, Inc. U.S. Environmental Protection Agency started working in photochemical modelling in 1970's, they have evolved an Empirical Kinetic Modelling Approach (EKMA), for studying the effect of change in organic and NO_x concentration on peak ozone level (Seinfeld, 1988). During early 1980's researchers felt the need to study the new sources of emission to represent more accurate scenario of air pollution, this led to the development of a

facility to estimate and control fugitive emissions. Fugitive sources include accidental release and leakages, evaporation from spills and waste treatment facilities and uncaptured process emissions. EPA has developed National Emission Standards for Hazardous Air Pollutants (NESHAPs), which included leak detection and repair (LDAR) program to tackle fugitive emissions. With the advancement in coal producing industries and power plants the focus of modelling agencies shifted to Sulphur dioxide and toxic pollutants. This phase began in 1990 where importance is given to large scale grid models (Irwin, 2004). The extent of modelling has been increased from this phase, from just showing compliance than to be used as an assessment tool for major pollutants. The attention towards radioactive and particulate modelling were drawn from atomic explosions which felt the need to develop models for assessing the emission released from nuclear radiations.

3. Methodology

The review presented in this paper compare different category of dispersion models, based on certain features in each category, Gaussian models - AERMOD, CALINE4, Airviro Gauss, CTDMPLUS and Fugitive Dust Model (FDM); Eulerian models – CALGRID, FARM, Panache, Airviro Grid; Lagrangian – GRAL, FLEXPART, AUSTAL2000, HYSPLIT and Advanced Dispersion Models - UK-ADMS5, TAPM and CALPUF. The work presented by different researcher available through literature has been used for the comparison purpose. Dispersion models provide means of estimating the concentration of pollutants in the atmosphere and require some specific input information about meteorology, emission source and topography. Dispersion models are the deterministic mathematical models which has been

categorized into steady state and time dependent models based on the approach they followed as illustrated in fig. 1.

Box models – These are the simplest dispersion models, which assumes pollutants inside the box representing a given volume of atmospheric air in a geographical region are homogeneously distributed inside the box.

Gaussian models – Gaussian Dispersion Models are based on determining the concentration of pollutant puff as it moves away from the source. A combination of infinitely rapid series of small individual puffs or plume of polluted air has been considered for modelling purpose.

Eulerian models – Eulerian model tracks the movement of polluted plume through fixed frame of reference.

Lagrangian models – Lagrangian models follow the movement of individual particle as it moves in time and space.

Advanced Dispersion Models – These models require three-dimensional meteorological fields and an assumption of spatial uniformity.

The comparison has been done based on the following features (Atmospheric dispersion models, EPA, 1993):

- **Geographic scale** – the geographical extent of dispersion models is considered based on pollution exposure as near field (≤ 50 km) and far field (> 50 km). Most of the dispersion models works in near field region as population are more exposed to pollution near emission source. But due to widespread emission and accumulation of

pollutants at low concentration, dispersion more than 50 km range are more of concern now.

- **Time scale** – Time scale represents the outcome of model obtained for a period of short-term exposure, i.e. 1 hour, 8 hours or 24-hours average concentration or annual average concentration for long term exposure.
- **Land use** – Land use pattern determine the dispersion coefficient. For single source models, modelling area is either rural or urban but for modelling multisource, land use can be rural or urban depends on most of the sources lie in that area.
- **Terrain** – Most of the dispersion models are suited for flat terrain but models accommodating complex terrain represent more realistic picture. EPA has modified several existing models which can be adapted to site specific complex scenarios.
- **Source configuration** – Models can be categorised based on the sources of pollution in four types as shown in fig. 2.
- **Form of release** – Models can be developed differently based on the form of pollutants, i.e. gaseous or particulates.
- **Removal Mechanism** – Atmospheric chemistry involves dissolution, adsorption, gravitational settling and precipitation which affects the removal mechanism of pollutants from the atmosphere. This factor is important in determining the efficiency of models.
- **Pre-Processor Unit** – Some models are employed with pre-processor units for providing input data for different parameters like meteorological, terrain etc.

- **Input parameters** – These are the input variable on which the processing of model depends. Some of the required input parameters for included dispersion models has been added in the tables 1 to 4. The parameters given does not include a complete data set because the actual data set depends on the usage and application of models.
- **Working mechanism** – This feature describes the approach followed by different models in accommodating the atmospheric physical processes and the dispersion of pollutants concentration vertically and horizontally.
- **Applications** – This feature includes various application associated with the models.

The comparison of different dispersion models in each mentioned category are presented in table 1 to table 4.

4. Limitations of dispersion models

Air quality models are built to present a real picture of pollution scenario but, they are based on the inadequate understanding of the complex atmospheric system involved in the dispersion and transformation of pollutants. Inaccuracies in emission inventories and uncertainties in the quality or representativeness of meteorological data are also major contributors to the overall inaccuracy of a model prediction (Good Practice Guide for Atmospheric Dispersion Modelling Ministry of Environment, New Zealand, 2004).

4.1 Gaussian models

1. Gaussian models do not consider the causality effect of wind as they are based on the assumption that, the pollutants are transported in a straight line like a beam towards

receptors at some distance away from the source.

2. These models are suitable for wind speed above 0.5 or 1 m/s. Speed lower than this, restrict their application.
3. These models are not suitable for moderate and complex terrain as they do not account for turning or rising wind caused by the terrain itself.
4. These models are not suitable for on-field site emission as they lack experimental data to calibrate dispersion coefficient for near field conditions (Atmospheric dispersion models, EPA, 1993).
5. The model does not have memory of the pollutants released during previous hour. This limitation is especially important for the proper simulation of morning inversion break-up, fumigation and diurnal recycling of pollutants over cities.

4.2 Advanced dispersion models including Lagrangian and Eulerian

1. These models require an exhaustive understanding of boundary-layer meteorology, atmospheric turbulence, mesoscale meteorology, atmospheric chemistry and particle dynamics.
2. A large amount of input parameters is required, which creates a complex user interface, which in-turn require post-processing software to handle large output files.
3. These models involve huge risk due to limited understanding and fewer experts of the field.
4. A fully three-dimensional, time-dependent meteorological data set is usually required, which needs a good understanding of air pollution meteorology.

5. Particle models are best in simulating pollutant concentration near to the source.

However, they need at least 105 particles to represent a pollutant release.

6. Puff models do not represent a realistic picture of pollutant distribution, although they are less expensive computationally.

5. Conclusion

Air quality models are the important tool for analysing the quality of air. They provide a relatable picture of existing conditions and future scenario for policy makers to develop and implement air management plan which can keep the pollution under control and provide a healthy atmosphere to live in. Although they are not accurate enough to simulate the exact atmospheric processes which governs the dispersion and transformation of pollutants, but they can be capable of producing close results. These results can be conveniently used by various stakeholders concerning the environment. Through this paper, the developmental phase in the field of air quality modelling with emphasis on dispersion modelling has been presented. The review highlighted that, the advancement in modelling studies had followed the need of the time from simple gaussian plume models to more advanced particle, grid and trajectory models based on Eulerian and Lagrangian approach. The comparison among different models in the same categories reveals the structure and specification of these models as well as their limitations and applications for different purpose (Table 1 to Table 4). Thus, since 1940's modelling field has advanced leaps and bounds and become more specific from general. Still the modelling arena has much to explore, with the new sources of pollution are creating, new pollutants are releasing, dynamic atmospheric processes demand new ways of handling

dispersion and transformation of pollutants.

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Table 1. Comparison of different Gaussian models

Features	AERMOD	CALINE4	Airviro Gauss	CTDMPLUS	FDM
Geographic scale	Near Field	Near Field	Near field (100 m to 2 km)	Near Field	Near Field
Time Scale	Long term exposure	Long term exposure	Short term exposure	Short term exposure	Short and long term both
Land Use	Urban and rural both	Urban and rural both	Urban	Urban/Rural	Urban and rural both
Terrain	Suitable for working in both simple and complex terrain	Simple Terrain	Simple/Flat terrain	Complex terrain	Simple terrain
Source Configuration	Point, area, and volume sources	Line source	Point, road, area and grid sources	Elevated point sources at receptors	Point, line and area sources
Form of release	Gaseous as well as particulates	Gaseous- CO, NO ₂ Particulates- suspended particles	Gaseous as well as particulates	Gaseous as well as particulates	Fugitive dust sources
Removal Mechanism	Accommodate dry or wet deposition	Accommodate no removal mechanism	Accommodate no removal mechanism	Accommodate no removal mechanism	1. Deposition algorithm 2. Turbulent and gravitational settling
Pre-Processor unit	AERMET - Meteorological preprocessor	No pre-processor unit is employed	Emission data is stored within the Emission	Pre-processor for terrain and	EPA RAMMET - Meteorological

	Database (EDB). Databases within the model allow it to store emission characteristics from an unlimited number of sources		meteorological parameters	preprocessor
	AERMAP - Terrain pre-processor			
	AERSURFACE - Geophysical parameters (roughness length, albedo, Bowen ratio)			
Input Parameters	• Meteorological parameters generated by AERMET – surface boundary layer parameters, vertical profile parameters like wind speed, wind direction and turbulence. • Terrain preprocessor generated by AERMAP – Hill height scales and receptor elevation. • Pollutant type, Averaging time, Dispersion co-efficient, Emission rate, no. and type of receptors.	• Meteorological parameters - wind speed, wind direction, stability class, ambient temperature, cloud base height, cloud cover, mixing height and standard deviation of wind direction. • Surface roughness height. • Discrete receptors	• Plume Length, stability class, roughness value, wind field and height above ground. Not much information is available in literature.	• Meteorological parameters generated by RAMMET- Wind speed, wind direction, atmospheric stability, temperature and, mixing height. • Deposition velocity and gravitational settling velocity (optional) or characteristic diameter of particle.

Working Mechanism	1. In stable boundary conditions concentration distribution both horizontally and vertically is assumed to be Gaussian.	1. Horizontal homogenous wind flow.	1. The Gauss model is based on a Lagrangian Gaussian formulation.	Uses probability distribution function to measure vertical distribution in convective boundary condition (unstable atmosphere).	1. Model incorporates a detailed deposition algorithm based on equation of Ermak.
	2. In convective boundary condition (unstable atmosphere) horizontal distribution is treated as gaussian while vertical distribution is non- gaussian (bi-Gaussian probability density function)	2. Steady state meteorological conditions.	2. Stability calculations are based on Monin-Obukhov length criteria.		2. Eddies diffusion is assumed as constant. 3. Emission rate is considered as a function of wind speed.
Applications	Capable of simulating many sources with different shapes, at ground or elevated, buoyant or non-buoyant, emitting one or more pollutants.	Prediction of air pollution concentration near roadways	1. This model is used to simulate ground concentrations distribution of pollutants over urban or industrial areas. 2. Accumulation of polluted air over a period of several days cannot be studied with this model	This model is used to assess stable and neutral atmospheric conditions as well as Daytime, unstable conditions.	This is a computerized model used for computing concentration and deposition impacts from fugitive dust sources.

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Developing Agency	AERMIC (American Meteorological Society (AMS)/United States Environmental Protection Agency (EPA) Regulatory Model Improvement Committee)	California department of transportation (Caltrans)	Swedish Meteorological and Hydrological Institute	Environmental Protection Agency (EPA)	TRC Environmental Consultants
Reference	Cimorelli et al., 2004	Benson, 1989	Airviro User's Reference, 2015	Peters et al., 1999	Winges, 1990

Table 2. Comparison of different Eulerian models

Features	REM-CALGRID	FARM	PANACHE
Geographic scale	Far Field	-	Near and medium field
Time Scale	Short term/Long term exposure	Short term/Long term exposure	Short term exposure
Land Use	Rural and Urban	Urban or regional scale	Industrial and urban
Terrain	Complex terrain	Simple and Complex terrain	Simple and Complex terrain
Source Configuration	Point, line, area and volume sources	Area and point sources	Stack, point, area and volume sources
Form of release	Oxidants and Aerosols	Gaseous pollutants, Persistent organic pollutants (POPs) and heavy metals (HM).	Toxic & hazardous, Chemically reactive & non-reactive gases and particulate emission.
Removal Mechanism	1. Dry deposition and Wet deposition 2. Gas-phase chemistry 3. Aerosol treatment	1. Dry removal of pollutants dependent on local meteorology and land-use. 2. Wet removal through precipitation scavenging processes. 3. Aerosol modules (AERO3) 4. Photochemistry	1. Scavenging 2. Acid rain 3. Evaporation 4. Chemical reactions with gases 5. Settling and dry deposition 6. Surface deposition of aerosol

Pre-Processor unit	Geophysical data preprocessors for extracting land-use and terrain data	Kinetic Pre-Processor for gas phase chemical reactions.	-
Input parameters	• Geographical parameters - Terrain elevation, roughness length, albedo, leaf area index and land cover type. • Time – independent data on chemistry parameter. • Boundary and initial concentration data. • Emission data • Meteorological data obtained from CALMET.	• Emission data • Meteorological data • Initial concentration • Land Cover data – Roughness length. • Chemical species characteristics.	• Terrain – Location, average ground roughness, altitude contours, terrain map showing building, arbitrary building, urban area, etc., urban area, average building height and anthropogenic heat flux. • Meteorology - Wind speed and direction, ambient air temperature, relative humidity, stability conditions, ceiling height and cloud cover. • Chemical Properties – Molecular weight, Specific internal energy and molecular diffusion coefficient.

Working Mechanism	1 Vertical advection is treated with a Crank-Nicolson-scheme. 2. Vertical turbulent mixing formulation uses K-diffusion. 3. Similarity theory for stable and convective boundary layer. 4. Diffusion coefficients based on Planetary boundary layer scaling regimes. 5. It uses mass continuity to track material throughout a gridded volume	1. Three-dimensional transport by advection and turbulent diffusion. 2. Physical and chemical process influencing the concentration fields within the modelling domain are described by a system of partial differential equations (PDE).	1. Computational fluid dynamics model (CFD). 2. Uses CFD tools (Naiver Stokes equations and turbulence models) in a finite volume-based approach.
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Applications	Modelling of pollutants (photo-oxidants, aerosols) on different scales.	Flexible Air Quality Regional (FARM) model may be applied to evaluate the impact of different emission scenarios, to forecast air quality levels and to produce air quality maps particularly useful at location where no monitoring stations are available.	PANACHE is a Eulerian (and Lagrangian for particulate matter), 3-dimensional finite volume fluid mechanics model designed to simulate continuous and short-term pollution dispersion in the Atmosphere. PANACHE may also be used in emergency planning and management for episodic emissions, and fire and soot spread in forested and urban areas or from combustible pools.
Developing Agency	California Air Resources Board (CARB)	ARIANET	ADEME (French Ministry and Environment Agency)
Reference	Stern et al., 2009	Calori et al., 2016	Transoft Group, 1994

Table 3. Comparison of different Lagrangian models

Features	GRAL	FLEXPART	AUSTAL2000	HYSPLIT
Geographic scale	Near Field	Far Field	Near/Far Field (upto 100 km)	Near/Far field
Time Scale	Short term exposure	Short/Long term exposure	Short/Long term exposure	Short/Long term exposure
Land Use	Street Canyons to Urban	Urban and Rural	Does not differentiate between urban and rural area.	Urban and Rural
Terrain	Flat and Complex terrain	Flat and complex terrain	Flat and complex terrain	Flat and complex terrain
Source Configuration	Line, point and area sources	Point, line, area or volume sources.	Point, line, area, and volume sources	Point, line and area sources
Form of release	Non-reactive pollutants, Gaseous and particulates.	Gases, Aerosols, black carbon and volcanic ash	Gaseous pollutants and Particulate matter (Smaller than 2.5 μ to larger than 50 μ m)	Gaseous pollutants and Particulate matter, allergens, volcanic ash and hazardous material.
Removal Mechanism	1. Dry and wet deposition	1. Dry and wet deposition	1. Atmospheric advection and turbulent mixing	1. Wet and dry deposition
	2. Sedimentation.	2. Radioactive decay	2. Dry and wet deposition, sedimentation	2. Radioactive decay, and Resuspension.
	3. Building downwash effects	3. Gravitational settling	3. Chemical conversion	
	4. Decay rates (e.g. bacteria die off, radioactive decay)	4. 1st-order chemical reactions	4. Cloud radiation	
			5. Gravitational settling of dust	

Pre-Processor unit	Micro-scale flow-field model	Dust mobilization scheme runs as a pre-processor.		
		FLEXTRA – a kinematic trajectory model manages input data.	-	-
Input Parameters	• Topography, Land–use type and buildings. • Roughness length, Heat capacity, Thermal conductivity, Albedo and Soil moisture. • Emission data – Total emission by vehicle category, heating systems and industry. • Meteorological data – Wind speed and direction, Temperature data.	• Horizontal and vertical wind components, temperature and specific humidity. • Surface pressure, Dew point temperature, Large scale and convective precipitation, Sensible heat flux, East/West and North/South Surface stress, Topography, Land-sea-mask and Subgrid standard deviation of topography and land-use inventory.	• Wind measurement height, wind direction, wind speed, and the stability classes according to Klug–Maniel (German standard stability transformation). • Roughness length and co-ordinates of area.	• HYSPLIT uses gridded meteorological data fields generated and archived from other meteorological models. • Stability and mixing are estimated from the meteorological input data if it contains heat and momentum fluxes otherwise from temperature and wind data.

Working Mechanism	1. The atmospheric stability is calculated using boundary layer scaling. 2. The model assumes a constant plume rise in the vicinity of the tunnel portal as a function of the temperature difference between the ambient air and the tunnel flow.	It considers the skewness in the vertical velocity distribution (updrafts and downdrafts) and vertical gradients in air density.	1. Stability calculations are based on Monin - Obukhov length criteria. 2. This model simulates the trajectories of particles rather than investigating the fluxes. This offers more precision in modeling physical process.	1. The dispersion of a pollutant is calculated by assuming either puff or particle dispersion. 2. The calculation of dispersion rate is done from vertical diffusivity profile, wind shear and horizontal deformation of the wind field. 3. Model is a hybrid between Lagrangian approach for advection and dispersion calculations and Eulerian approach for pollutant concentration calculations.
Applications	1. The Graz Lagrangian Model can handle steady state (standard mode) as well as transient simulations (e.g. puff releases) 2. GRAL is applied for the investigation of the impact of different traffic emission.	1. This model has great advantage of running in forward or backward mode. The forward mode is typically used to determine the downwind concentration or mixing ratio of pollutants. The backward mode can be used to determine the origin	1. AUSTAL2000 is a reference implementation program developed on behalf of the German Federal Environmental Agency. 2. Regulatory Practice 3. Odour assessment	1. It is used to compute air parcel trajectories and deposition or dispersion of atmospheric pollutants. 2. It is used to establish whether high levels of air pollution at one location are caused by transport of air

	3. It involves special algorithms to deal with dispersion from tunnel portals and under conditions of low wind speed.	of observed emissions.		contaminants from another location.
		2. FLEXPART has been used for inverse modeling based on source–receptor relationships.		3. It is used for tracking and forecasting the release of radioactive material, wildfire smoke, windblown dust and pollutants.
				4. It can be used for both research and emergency response applications.
Developing Agency	Institute for Internal Combustion Engines and Thermodynamics of TU Graz	Norwegian Institute for Air Research.	Federal Environmental Agency (UBA), Germany	National Oceanic and Atmospheric Administration, and Australia's Bureau of Meteorology.
Reference	Holmes and Morawska, 2006	Pisso et al., 2019	AUSTAL2000 Program Documentation of Version 2.6, 2014	Stein et al., 2015

Table 4. Comparison of different advanced dispersion models

Features	ADMS 5	TAPM	CALPUFF
Geographic scale	Applicable up to 60 km downwind of the source	Near Field	Applicable for both Near Field and Far Field range
Time Scale	Short term/Long term exposure	Long term exposure	Short term/Long term exposure
Land Use	Urban	Rural and urban both	Rural and urban both
Terrain	Complex terrain	Suitable for working in both simple and complex terrain	Rough and complex terrain
Source Configuration	Point, jet (directional release), line, area and volume sources.	point, line, area and volume sources	point, line, area and volume sources
Form of release	Gaseous as well as particulates	Gaseous as well as particulates	Reactive/Non-reactive gaseous pollutants as well as particulates.
Removal Mechanism	1. Accommodate dry or wet deposition	1. Plume rise and building wake effects	1. Chemical removal
	2. NO _x chemistry using a simple reaction scheme for NO _x , NO ₂ and ozone	2. Wet and dry deposition	2. Wet and dry deposition
	3. Impacts of hills	3. Chemistry module for urban air-shed applications	3. Building downwash
Pre-Processor unit	In-built meteorological pre-processor	Pre-processor for providing meteorological data inputs.	4. Plume fumigation
			CALMET - 3-dimensional meteorological model Pre-processor for geophysical datasets

Input Parameters	• Source data: location, emission rate, physical stack height, stack gas exit velocity, stack inside diameter, and stack gas temperature. • Operational inputs include source elevation, building dimensions, particle size distribution with corresponding settling velocities, and surface reflection coefficients. • Meteorological data: hourly surface observations for input into the meteorological pre-processor or boundary layer parameters (boundary layer height, Monin - Obukhov length, surface heat flux, etc).	• This model follows a prognostic approach. Its air pollution component uses the model-predicted three-dimensional meteorology and turbulence. • Land-use data and Refinery generated heat flux. • Topography, Monthly sea-surface temperature, Soil types, Deep soil moisture content, and Deep soil temperature.	CALPOST – A post processing package • Time-varying emissions, Complex terrain hill data, Turbulence data, Chemical transformation rates, Deposition velocities (for deposition only), and hourly ozone data. • Hourly surface observations. • Upper air data. • Hourly precipitation data. • Geophysical parameters -Land use, Terrain heights, Surface roughness length, Albedo, Bowen ratio, Soil and Anthropogenic heat fluxes, and Leaf area index.
Working Mechanism	1. Atmospheric boundary layer properties are characterized by two parameters: the boundary layer depth, and the Monin-Obukhov length 2. A skewed Gaussian concentration distribution to calculate dispersion under convective conditions.	1. Mathematical formulation follows semi-empirical/analytic approaches. 2. The meteorological component of TAPM is an incompressible, non-hydrostatic, primitive equation model with a terrain-following vertical coordinate for three-dimensional	1. Pasquill - Grifford dispersion co-efficient (rural) 2. McElory-Pooler dispersion co-efficient (urban) 3. Puff splitting and Differential advection & dispersion.

Applications	1. Assessment of pollutant concentrations against air quality standards.	simulations.	
	2. Estimation for radioactive decay including γ -ray dose.	1. TAPM is a model developed to estimate the spread and impact of air pollution.	1. CALPUFF is a non-steady state Lagrangian Gaussian puff model adopted by the United States Environmental Protection Agency (EPA) as a preferred model for assessing long range transport of pollutants and their impacts on Federal Class I areas and on a case-by-case basis for certain near-field applications involving complex meteorological conditions.
	3. Odour modelling,	2. Prognostic meteorological and air pollution concentration components, eliminates the need to have site-specific meteorological observations.	
	4. Estimation of emissions around buildings.		
	5. Dispersion in coastal areas.		
	6. Used to model time varying emissions.		2. The modelling system has a capability of interfacing to several other models.
Developing Agency	Cambridge Environmental Research consultants (CERC)	Australia's Commonwealth Scientific and Industrial Research Organization (CSIRO)	Exponent, Inc.
Reference	ADMS5- User Guide, 2016	Hurley, 2008	Scire et al., (2000)

Figure 1. Schematic representation of types of dispersion models

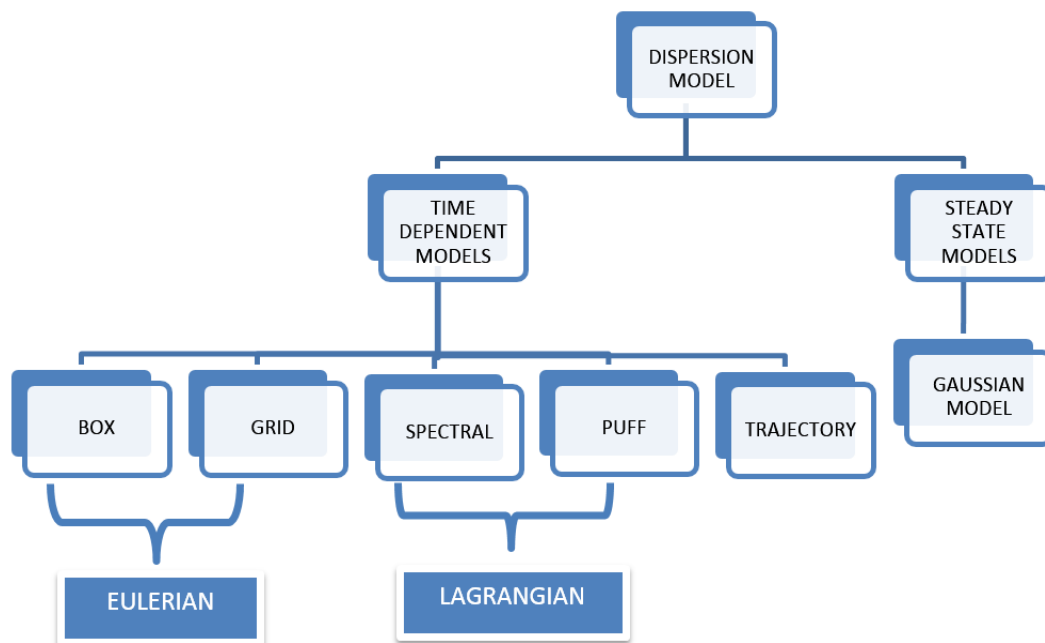


Figure 2. Type of dispersion sources

